$\boxed{\text{Lecture 4}}$ $\boxed{5.07.11}$

Equations of motion for the GFs

1. Our

$$\hat{\mathcal{H}} = \int dx\, \psi^\dagger(x)\, h(xt)\, \psi(x) + \frac{1}{2}\int dx\,dx'\, \psi^\dagger(x)\psi^\dagger(x')\, v(x,x')\, \psi(x')\psi(x) \qquad (1)$$

$$h(xt) = -\frac{1}{2}\Delta_x + \underbrace{U(xt)}_{\text{external field}} - \mu \qquad \left[\vec{A}(rt) \text{ can also be included if desired}\right]$$

In the GF we have $\psi_{\mathcal{H}}(xt) = U(t,t_0)\,\psi(x)\,U(t_0,t)$, and hence for the EoM we need to differentiate those. For any $A_{\mathcal{H}}(x,t)$:

$$i\partial_t A_{\mathcal{H}}(xt) = i\partial_t\, U(t,t_0)\, A(x)\, U(t_0,t) = \left(i\partial_t U(t,t_0)\right) A(x)\, U(t_0,t)$$

$$+\, U(t,t_0)\, A(x)\left(i\partial_t\, U(t_0,t)\right) = \mathcal{H}_t U(t,t_0)\, A\, U(t_0,t) - U(t,t_0)\, A\, U(t_0,t)\mathcal{H}_t$$

$$= [\mathcal{H}_t, A_{\mathcal{H}}(xt)] \qquad (2)$$

But

$$[A_{\mathcal{H}}, B_{\mathcal{H}}] = A_{\mathcal{H}}B_{\mathcal{H}} - B_{\mathcal{H}}A_{\mathcal{H}} = U(t_0,t)\, A\, \overbrace{U(t,t_0)\cdot U(t,t_0)}^{1}\, B\, U(t_0,t) - \dots$$

$$= U(t_0,t)\, [A,B]\, U(t,t_0) \equiv [A,B]_{\mathcal{H}} \qquad (3)$$

so that we can calculate first the usual commutators and then convert them into the Heisenberg representation.

- $$[\psi(x), \underbrace{\int dy\, \psi^\dagger(y)\, h(y)\, \psi(y)}_{\mathcal{H}_1}] = \int dy\, [\psi(x), \psi^\dagger(y)\, h(y')\, \psi(y')]_{y'\to y}$$

$$= \int dy \left(h(y')\, [\psi(x), \psi^\dagger(y)\psi(y')]\right)_{y'\to y} \qquad (4)$$

- Properties of the field operators: anticommutator

$$\{\psi(x), \psi(x')\} = \sum_{nm} \psi_n(x)\psi_m(x')\, \{a_n, a_m\} = 0 \qquad (5)$$

$$\{\psi^\dagger(x), \psi^\dagger(x')\} = 0 \qquad (6)$$

complete set

$$\{\psi(x), \psi^\dagger(x')\} = \sum_{nm} \psi_n(x)\psi_m^*(x')\, \underbrace{\{a_n, a_m^\dagger\}}_{\delta_{nm}} = \sum_n \psi_n(x)\psi_n^*(x') = \delta(x-x') \qquad (7)$$

• Therefore:

$$[\psi(x), \psi^\dagger(y)\psi(y')] = \psi_x \psi^\dagger_y \psi_{y'} - \psi^\dagger_y \overbrace{\psi_{y'} \psi_x}^{-\psi_x\psi_{y'}} = \overbrace{\psi_x \psi^\dagger_y}^{-\psi_x\psi^\dagger_y + \delta(x-y)} \psi_{y'} + [\psi^\dagger_y \psi_x]\, \psi_{y'} = \psi(y')\,\delta(x-y),$$

$$\hookrightarrow [\psi(x), \hat{\mathcal{H}}_1] = \int dy \left[h(y')\,\psi(y')\,\delta(x-y)\right]_{y' \to y}$$

$$= \int dy\, \delta(x-y)\, h(y)\,\psi(y) = h(x)\,\psi(x) \qquad (8)$$

• Now a similar calculation for the 2-particle part $\mathcal{H}_2$:

$$[\psi(x), \mathcal{H}_2] = \frac{1}{2}\int dx_1 dx_2\, v(x_1,x_2)\, [\psi_x, \psi^\dagger_{x_1}\psi^\dagger_{x_2}\psi_{x_2}\psi_{x_1}]$$

$$[\psi_x, \psi^\dagger_{x_1}\psi^\dagger_{x_2}\psi_{x_2}\psi_{x_1}] = [\psi_3, \psi^\dagger_1\psi^\dagger_2\psi_2\psi_1] = \underbrace{\psi_3\psi^\dagger_1}\psi^\dagger_2\psi_2\psi_1 - \psi^\dagger_1\psi^\dagger_2\psi_2\psi_1\psi_3$$

$$= (\delta_{13} - \psi^\dagger_1\psi_3)\psi^\dagger_2\psi_2\psi_1 - (2nd) = \delta_{13}\psi^\dagger_2\psi_2\psi_1 - \psi^\dagger_1\underbrace{\psi_3\psi^\dagger_2}\psi_2\psi_1 - (2nd)$$

$$= \delta_{13}\psi^\dagger_2\psi_2\psi_1 - \psi^\dagger_1(\delta_{32} - \psi^\dagger_2\psi_3)\psi_2\psi_1 - (2nd) = \delta_{13}\psi^\dagger_2\psi_2\psi_1 - \delta_{32}\psi^\dagger_1\psi_2\psi_1$$

$$+ \psi^\dagger_1\psi^\dagger_2\underbrace{\psi_3\psi_2\psi_1}_{=+\psi_2\psi_1\psi_3} - \psi^\dagger_1\psi^\dagger_2\psi_2\psi_1\psi_3 \equiv \delta_{13}\psi^\dagger_2\psi_2\psi_1 - \delta_{23}\psi^\dagger_1\psi_2\psi_1$$

$$\hookrightarrow [\psi_x, \mathcal{H}_2] = \frac{1}{2}\int dx_1 dx_2\, v(x_1,x_2)\left[\delta(x_1-x)\,\psi^\dagger_{x_2}\psi_{x_2}\psi_{x_1} - \delta(x_2-x)\,\psi^\dagger_{x_1}\psi_{x_2}\psi_{x_1}\right]$$

$$= \frac{1}{2}\int dx_2\, v(x,x_2)\,\psi^\dagger_{x_2}\psi_{x_2}\psi_x - \frac{1}{2}\int dx_1\, v(x_1,x)\,\psi^\dagger_{x_1}\psi_x\psi_{x_1}$$

permute these two

$$= \int dx'\, v(x,x')\,\psi^\dagger_{x'}\psi_{x'}\psi_x \equiv \int dx'\, v(x,x')\,\rho(x')\,\psi_x \qquad (9)$$

So, finally,

$x_1 t_1$

$$[\mathcal{H}_1, \psi_{\mathcal{H}}(1)] = \mathcal{U}(t_0,t_1)\,[\mathcal{H}_1, \psi(x_1)]\,\mathcal{U}(t_1,t_0)$$

$$= h(x_1)\,\mathcal{U}(t_0,t_1)\,\psi(x_1)\,\mathcal{U}(t_1,t_0) \equiv h(x_1)\,\psi_{\mathcal{H}}(1) \qquad (10)$$

and

$$[\mathcal{H}_2, \psi_{\mathcal{H}}(1)] = \mathcal{U}(t_0,t_1) \int dx'\, v(x_1,x')\, \psi^{\dagger}_{x'} \psi_{x'} \psi_{x_1}\, \mathcal{U}(t_1,t_0)$$

$$= \underbrace{\int dt_2 \int dx'}_{d2} v(x_1,x')\, \mathcal{U}(t_0,t_2) \psi^{\dagger}_{x'} \underbrace{\mathcal{U}(t_2,t_0)\, \mathcal{U}(t_0,t_2)}_{=1} \psi_{x'} \underbrace{\mathcal{U}(t_2,t_0)\, \mathcal{U}(t_0,t_1)}_{=1 \text{ due to } \delta(t_1,t_2)} \psi_{x_1} \mathcal{U}(t_1,t_0)\, \delta(t_1,t_2)$$

The $\delta(t_1,t_2)$ is incorporated into $v(x_1,x')$ as $v(1,2)$ ↓

$$[\mathcal{H}_2, \psi_{\mathcal{H}}(1)] = \int d2\, v(1,2)\, \psi^{\dagger}_{\mathcal{H}}(2) \psi_{\mathcal{H}}(2)\, \psi_{\mathcal{H}}(1) \qquad (11)$$

- Finally, we can write:

$$\boxed{i\partial_{t_1} \psi_{\mathcal{H}}(1) = h(1)\, \psi_{\mathcal{H}}(1) + \int d2\, v(1,2)\, \psi^{\dagger}_{\mathcal{H}}(2) \psi_{\mathcal{H}}(2)\, \psi_{\mathcal{H}}(1)} \qquad (12)$$

$$\boxed{v(1,2) = v(x_1,x_2)\, \delta(t_1,t_2)} \qquad (13)$$

- Similarly we calculate the commutators with $\psi^{\dagger}(x_1)$:

$$[\psi^{\dagger}(x_1), \mathcal{H}_1] = \int dy \Big[h(y')\Big(\psi^{\dagger}_{x_1} \psi^{\dagger}_{y} \psi_{y'} - \psi^{\dagger}_{y} \underbrace{\psi_{y'} \psi^{\dagger}_{x_1}}_{\delta_{x_1 y'} - \psi^{\dagger}_{x_1}\psi_{y'}} \Big)\Big]_{y' \to y}$$

$$= \int dy\, h(y') \Big(\psi^{\dagger}_{x_1} \psi^{\dagger}_{y} \psi_{y'} - \delta_{x_1 y'} \psi^{\dagger}_{y} + \underbrace{\psi^{\dagger}_{y} \psi^{\dagger}_{x_1}}_{\text{permute}} \psi_{y'} \Big)_{y' \to y} = -h(x_1)\, \psi^{\dagger}(x_1),$$

$$[\psi^{\dagger}_{x_3}, \mathcal{H}_2] = \frac{1}{2} \int dx_1 dx_2\, v(x_1,x_2)\, [\psi^{\dagger}_{x_3}, \psi^{\dagger}_{x_1} \psi^{\dagger}_{x_2} \psi_{x_2} \psi_{x_1}] =$$

$$= \frac{1}{2} \int dx_1 dx_2\, v(x_1,x_2) \Big(\delta_{32} \psi^{\dagger}_1 \psi^{\dagger}_2 \psi_1 - \delta_{13} \psi^{\dagger}_1 \psi^{\dagger}_2 \psi_2 \Big)$$

$$= -\int dx_1\, v(x_1,x_3)\, \psi^{\dagger}_{x_3} \psi^{\dagger}_{x_1} \psi_{x_1} = -\int d1\, v(1,3)\, \psi^{\dagger}(3) \psi^{\dagger}(1) \psi(1)$$

↳
$$\boxed{i\partial_{t_1} \psi^{\dagger}_{\mathcal{H}}(1) = -h(1)\, \psi^{\dagger}_{\mathcal{H}}(1) - \int d2\, v(1,2)\, \psi^{\dagger}_{\mathcal{H}}(1) \psi^{\dagger}_{\mathcal{H}}(2) \psi_{\mathcal{H}}(2)} \qquad (14)$$

2. EoM for the GF

- $$G(1,2) = \theta(1,2)\, \underbrace{G^{>}(1,2)}_{-i\langle \psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)\rangle} + \theta(2,1)\, \underbrace{G^{<}(1,2)}_{+i\langle \psi^{\dagger}_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)\rangle} \qquad (15)$$

$\hookrightarrow$ $$i\partial_{t_1} G(1,2) = \left(i\partial_{t_1}\theta(t_1,t_2)\right)(-i)\langle \psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)\rangle + \theta(1,2)(-i)\left\langle\left(i\partial_{t_1}\psi_{\mathcal{H}}(1)\right)\psi^{\dagger}_{\mathcal{H}}(2)\right\rangle$$

$$+\left(i\partial_{t_1}\theta(t_2,t_1)\right) i\langle \psi^{\dagger}_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)\rangle + \theta(2,1)\, i\left\langle \psi^{\dagger}_{\mathcal{H}}(2)\left(i\partial_{t_1}\psi_{\mathcal{H}}(1)\right)\right\rangle$$

Here: $$\partial_{t_1}\theta(t_1,t_2) = \delta(t_1-t_2) \quad , \quad \partial_{t_1}\theta(t_2,t_1) = -\delta(t_1,t_2) \qquad (16)$$

- Two terms with these derivatives combine into:

$$+\delta(t_1-t_2)\langle \psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)\rangle + \delta(t_2-t_1)\langle \psi^{\dagger}_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)\rangle$$

$$= \delta(t_1-t_2)\langle \underbrace{\psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2) + \psi^{\dagger}_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)}\rangle$$

times are the same!

$$= \delta(t_1-t_2)\langle U(t_0,t_1)\left(\underbrace{\psi(x_1)\psi^{\dagger}(x_2) + \psi^{\dagger}(x_2)\psi(x_1)}_{\delta(x_1-x_2)}\right)U(t_1,t_0)\rangle$$

$$= \delta(t_1-t_2)\delta(x_1-x_2)\underbrace{\langle \underbrace{U(t_0,t_1)U(t_1,t_0)}_{=1}\rangle}_{=1} = \delta(1,2) \qquad (17)$$

- The other 2 terms:

$$-i\theta_{12}\left\langle\left(i\partial_{t_1}\psi_{\mathcal{H}}(1)\right)\psi^{\dagger}_{\mathcal{H}}(2)\right\rangle = -i\theta_{12}\Big[h(1)\langle \psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)\rangle +$$

$$+\int d3\, v(1,3)\langle \psi^{\dagger}_{\mathcal{H}}(3)\psi_{\mathcal{H}}(3)\psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)\rangle\Big] \qquad (18)$$

and

$$i\theta_{21}\langle\psi^\dagger_{\mathcal{H}}(2)\left(i\partial_{t_1}\psi_{\mathcal{H}}(1)\right)\rangle = i\theta_{21}\Big[h(1)\langle\psi^\dagger_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)\rangle +$$

$$+\int d3\, v(1,3)\langle\psi^\dagger_{\mathcal{H}}(2)\psi^\dagger_{\mathcal{H}}(3)\psi_{\mathcal{H}}(3)\psi_{\mathcal{H}}(1)\rangle\Big] \qquad (19)$$

Combine all 3 results (17), (18), (19):

$$i\partial_{t_1} G(1,2) = \delta(1,2) + h(1)\underbrace{\Big[\theta_{12}(-i)\langle\psi_{\mathcal{H}}(1)\psi^\dagger_{\mathcal{H}}(2)\rangle + \theta_{21}(+i)\langle\psi^\dagger_{\mathcal{H}}(2)\psi_{\mathcal{H}}(1)\rangle\Big]}_{G(1,2)}$$

$$+\int d3\, v(1,3)\Big\{\theta_{12}(-i)\langle\overbrace{\psi^\dagger_{\mathcal{H}}(3)\psi_{\mathcal{H}}(3)}^{\rho(3)}\psi_{\mathcal{H}}(1)\psi^\dagger_{\mathcal{H}}(2)\rangle$$

$$+\theta_{21}(+i)\langle\psi^\dagger_{\mathcal{H}}(2)\underbrace{\psi^\dagger_{\mathcal{H}}(3)\psi_{\mathcal{H}}(3)}_{\rho(3)}\psi_{\mathcal{H}}(1)\rangle\Big\}$$

$$= \delta(1,2) + h(1)G(1,2) + \int d3\, v(1,3)(-i)\Big[\theta_{12}\langle\rho(3)\psi_1\psi_2^\dagger\rangle - \theta_{21}\langle\psi_2^\dagger\rho_3\psi_1\rangle\Big] \qquad (19)^*$$

In $[\ldots]$: $\rho(3)$ is always on the left of ψ_1, so time $t_3 > t_1$ must be.
In $v(1,3)$ $t_1 = t_3$, but we can use $t_3^+ \Rightarrow t_3 + \delta$ instead of t_3 which then would make it "later" (δ is on the contour C, so it can be either positive, negative or pure imaginary). Then:

$$[\ldots] = \langle T_C\, \rho(3^+)\psi_1\psi_2^\dagger\rangle$$

Alternatively, we make t_1 slightly ~~larger~~ ~~smaller~~ "smaller" than t_3:

$$\int d3\, v(1,3)\ldots \Rightarrow \int d3\, v(1^+,3)(-i)\langle T_C(\psi_1\psi_2^\dagger\rho_3)\rangle$$

$\rho_3 = \psi_3^\dagger\psi_3$ contains two operators, no change of sign!

Indeed, if $t_1 > t_2$, then $t_3 > t_2$, and

$T_C(\psi_1\psi_2^\dagger\rho_3) = \rho_3\psi_1\psi_2^\dagger$ as it should;

if $t_1 < t_2$, then $t_3 < t_2 \rightarrow T_C(\psi_1\psi_2^\dagger\rho_3) = -\psi_2^\dagger\rho_3\psi_1$ — " —

$$\boxed{\left(i\partial_{t_1}-h_1\right) G(1,2) = \delta_{12} - i\int d3\, v(1,3^+) \langle T_C(\psi_1 \psi_2^\dagger \psi_3^\dagger \psi_3)\rangle} \quad (20)$$

3. Two-particle GF

$$\boxed{G_2(1,2,3,4) = \langle T_C\, \psi_{\mathcal{H}}(1)\, \psi_{\mathcal{H}}(2)\, \psi_{\mathcal{H}}^\dagger(3)\, \psi_{\mathcal{H}}^\dagger(4)\rangle} \quad (21)$$

↳ the integral term can be written via it:

$$\langle T_C(\psi_1 \psi_2^\dagger \psi_3^\dagger \psi_3)\rangle = \langle T_C(\psi_1 \psi_3 \underbrace{\psi_{3^+}^\dagger}_{} \psi_2^\dagger)\rangle \equiv G_2(1,3,3^+,2) \quad (22)$$

this will make $\psi_3^\dagger$ to appear before ψ_3

$$\boxed{\left(i\partial_{t_1}-h_1\right) G(1,2) = \delta_{12} - i\int d3\, v(1^+,3)\, G_2(1,3,3^+,2)} \quad (23)$$

4. Similarly

$$i\partial_{t_2} G(1,2) = i\partial_{t_2}\left[\theta_{12} G^>_{12} + \theta_{21} G^<_{12}\right] = i\left[-\delta_{12} G^>_{12} + \delta_{12} G^<_{12}\right] +$$

$$+\theta_{12}\left(i\partial_{t_2} G^>_{12}\right) + \theta_{21}\left(i\partial_{t_2} G^<_{12}\right) \qquad -\delta_{12}(G^> - G^<) \equiv -\delta_{12}\underbrace{\langle\{\psi_1\psi_2^\dagger + \psi_2^\dagger\psi_1\}\rangle}_{\delta(x_1-x_2)}$$

$$= -\delta_{12} - h(2)\, G(1,2) + i\int d3\, v(2,3^+) \langle T_C(\psi_{\mathcal{H}}(1)\, \psi_{\mathcal{H}}^\dagger(2)\, \rho_{\mathcal{H}}(3))\rangle$$

or

$$\boxed{\left(-i\partial_{t_2} - h(2)\right) G(1,2) = \delta(1,2) - i\int d3\, v(2,3^+) \underbrace{\langle T_C(\psi_1 \psi_2^\dagger \psi_3^\dagger \psi_3)\rangle}_{G_2(1,3^+,3^+,2)}} \quad (24)$$

5. EoM for G_2 involves G_3 and G_1; EoM for G_3 involves G_4 and G_2, etc. → chain of equations (infinite). In practice the chain is truncated by expressing G_n via G_m, $m < n$.

5. Self-energy

$$\boxed{-i\int d3\, v(1^+,3)\, G_2(1,3,3^+,2) \overset{\text{Def}}{=} \int d3\, \Sigma(1,3)\, G(3,2)} \quad (25)$$

and

$$\boxed{-i\int d3\, v(2,3^+)\, G_2(1,3,3^+,2) \overset{\text{Def}}{=} \int d3\, G(1,3)\, \tilde{\Sigma}(3,2)} \quad (26)$$

We prove that $\boxed{\Sigma(1,2) = \tilde{\Sigma}(1,2)}$ (27)

Proof (van Leeuwen)

(i) EoM for $\psi_{\mathcal{H}}(1)$ and $\psi^\dagger_{\mathcal{H}}(1)$:

$$\begin{cases} i\partial_{t_1}\psi_{\mathcal{H}}(1) = h(1)\psi_{\mathcal{H}}(1) + J_{\mathcal{H}}(1) & (28) \\ i\partial_{t_1}\psi^\dagger_{\mathcal{H}}(1) = -h(1)\psi^\dagger_{\mathcal{H}}(1) - J^\dagger_{\mathcal{H}}(1) & (29) \end{cases}$$

with

$$\begin{cases} J_{\mathcal{H}}(1) = \int d3\, v_{13}\psi^\dagger_3\psi_3\psi_1 = \int d3\, v_{13}\rho_3\psi_1 & (30) \\ J^\dagger_{\mathcal{H}}(1) = \int d3\, v_{13}\psi^\dagger_3\rho_3 \equiv \left(\int d3\, v_{13}\rho_3\psi_1\right)^\dagger \equiv \left(J_{\mathcal{H}}(1)\right)^\dagger & (31) \end{cases}$$

We shall omit the index $\mathcal{H}$ in what follows for simplicity of notations.

$$\begin{cases} (i\partial_{t_1} - h_1)\, G_{12} = \delta_{12} - i\langle T_C J_1\psi^\dagger_2\rangle \quad (\text{see } (19)^*) & (31) \\ (-i\partial_{t_2} - h_2)\, G_{12} = \delta_{12} - i\langle T_C \psi_1 J^\dagger_2\rangle \quad (\text{from } (24)) & (32) \end{cases}$$

$$\hookrightarrow \begin{cases} -i\langle T_C J_1\psi^\dagger_2\rangle \equiv \int d3\, \Sigma_{13} G_{32} & (33) \\ -i\langle T_C \psi_1 J^\dagger_2\rangle \equiv \int d3\, G_{13}\Sigma_{32} & (34) \end{cases}$$

ii) Apply $(-i\partial_{t_2} - h_2)$ to (33):

$$\text{RHS: } \int d3\, \Sigma_{13}\, (-i\partial_{t_2} - h_2)\, G_{32} = \int d3\, \Sigma_{13} \Big[\delta_{32} - i \underbrace{\langle T_C \psi_3 \mathcal{J}_2^\dagger \rangle}_{\int G_{34} \tilde{\Sigma}_{42}\, d4} \Big]$$

$$= \Sigma_{12} + \int d3\, d4\; \Sigma_{13}\, G_{34}\, \tilde{\Sigma}_{42}$$

~~LHS: = $i \langle T_C \mathcal{J}_1 (-i\partial_{t_2} - h_2) \psi_2^\dagger \rangle =$ = $i \langle T_C \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle$ = $i \langle T_C \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle$~~

LHS:

$$= (-i\partial_{t_2} - h_2)(-i) \langle T_C \mathcal{J}_1 \psi_2^\dagger \rangle = (-i\partial_{t_2} - h_2)(-i) \Big[\theta_{12} \langle \mathcal{J}_1 \psi_2^\dagger \rangle -$$

$$- \theta_{21} \langle \psi_2^\dagger \mathcal{J}_1 \rangle \Big] = - \underbrace{\partial_{t_2}}_{\text{applied only to } \theta\text{'s}} \Big[\theta_{12} \langle \mathcal{J}_1 \psi_2^\dagger \rangle - \theta_{21} \langle \psi_2^\dagger \mathcal{J}_1 \rangle \Big] +$$

$$+ (-i) \Big[\theta_{12} \langle \mathcal{J}_1 (-i\partial_{t_2} - h_2) \psi_2^\dagger \rangle - \theta_{21} \langle \big((-i\partial_{t_2} - h_2) \psi_2^\dagger \big) \mathcal{J}_1 \rangle \Big]$$

$$= - \Big[\underbrace{- \delta_{12} \langle \mathcal{J}_1 \psi_2^\dagger \rangle - \delta_{21} \langle \psi_2^\dagger \mathcal{J}_1 \rangle}_{- \delta_{12} \langle \{ \mathcal{J}_1, \psi_1^\dagger \} \rangle} \Big] + (-i) \Big[\theta_{12} \langle \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle - \theta_{21} \langle \mathcal{J}_2^\dagger \mathcal{J}_1 \rangle \Big]$$

$$= \delta_{12} \langle \{ \mathcal{J}_1, \psi_2^\dagger \} \rangle + (-i) \langle T_C \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle$$

$$\hookrightarrow \Sigma_{12} + \int d(34)\, \Sigma_{13}\, G_{34}\, \tilde{\Sigma}_{42} \equiv \delta_{12} \langle \{ \mathcal{J}_1, \psi_2^\dagger \} \rangle - i \langle T_C \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle \quad (35)$$

Similarly from (34) after acting with $(i\partial_{t_1} - h_1)$:

$$\tilde{\Sigma}_{12} + \int d(34)\, \Sigma_{13}\, G_{34}\, \tilde{\Sigma}_{42} = \delta_{12} \langle \{ \mathcal{J}_2^\dagger, \psi_1 \} \rangle - i \langle T_C \mathcal{J}_1 \mathcal{J}_2^\dagger \rangle \quad (36)$$

(here $\delta_{12} = \delta(t_1 - t_2)$, only wrt times! $x_1 \neq x_2$!!!)

If we prove that $\langle \{ \mathcal{J}_1, \psi_2^\dagger \} \rangle = \langle \{ \mathcal{J}_2^\dagger, \psi_1 \} \rangle$, then (27) will follow.

(iii) $\langle \{ \mathcal{J}_1, \psi_2^\dagger \} \rangle = \int d3\, v_{13} \langle \{ \rho_3 \psi_1, \psi_2^\dagger \} \rangle = \int d3\, v_{13} \langle \rho_3 \psi_1 \psi_2^\dagger + \psi_2^\dagger \rho_3 \psi_1 \rangle$

$$= \int d3\, v_{13} \left\langle U_{t_0 t_1} \underbrace{\rho_3 \psi_1 \psi_2^\dagger}_{\text{ordinary operators}} U_{t_1 t_0} + U_{t_0 t_1} \underbrace{\psi_2^\dagger \rho_3 \psi_1}_{\text{ordinary operators}} U_{t_1 t_0} \right\rangle$$

because $t_1 = t_3$ due to definition of v_{13}.

$$\hookrightarrow \langle \{ J_1, \psi_2^\dagger \} \rangle = \int d3\, v_{13} \langle U_{t_0 t_1} \underbrace{\{ \rho_3 \psi_1, \psi_2^\dagger \}}_{\text{ordinary operators.}} U_{t_0, t_1} \rangle$$

We get:

$$\{ \rho_3 \psi_1, \psi_2^\dagger \} = \psi_3^\dagger \psi_3 \psi_1 \psi_2^\dagger + \psi_2^\dagger \psi_3^\dagger \psi_3 \psi_1 = \left(\text{use anticommutation relations} \right)$$

$$= \delta_{12} \rho_3 - \delta_{23} \psi_3^\dagger \psi_1$$

$x_1 = x_2$ or $x_2 = x_3$ here

$$\hookrightarrow \langle \{ J_1, \psi_2^\dagger \} \rangle = \int d3\, v_{13} \langle U_{t_0 t_1} \left(\delta_{x_1 x_2} \rho_{x_3} - \delta_{x_2 x_3} \psi_{x_3}^\dagger \psi_{x_1} \right) U_{t_1 t_0} \rangle$$

$$= \delta_{x_1 x_2} \int d3\, v_{13} \langle \rho_3 \rangle - v_{12} \langle \psi_2^\dagger \psi_1 \rangle \qquad (37)$$

(but all times are identical to t_1)

Similarly,

$\langle \{ J_2^\dagger, \psi_1 \} \rangle =$ works out to be exactly the same!

That proves (27).

- Therefore, one can write:

$$\boxed{\begin{aligned} \left(i\partial_{t_1} - h_1 \right) G_{12} &= \delta_{12} + \int d3\, \Sigma_{13} G_{32} \\ \left(-i\partial_{t_2} - h_2 \right) G_{12} &= \delta_{12} + \int d3\, G_{13} \Sigma_{32} \end{aligned}} \qquad (38)$$

6. Look at (35) or (36):

$$\Sigma_{12} = \underbrace{\delta_{t_1 t_2} \langle \{ J_1, \psi_2^\dagger \} \rangle}_{\delta_{t_1 t_2} \Sigma^\delta_{12}} - \underbrace{\int d(34)\, \Sigma_{13} G_{34} \Sigma_{42} - i \langle T_C J_1 J_2^\dagger \rangle}_{\theta_{t_1 t_2} \Sigma^>_{12} + \theta_{t_2 t_1} \Sigma^<_{12}}$$

$$\hookrightarrow \boxed{\Sigma(1,2) = \delta_{t_1 t_2} \Sigma^\delta(1,2) + \theta_{t_1 t_2} \Sigma^>(1,2) + \theta_{t_2 t_1} \Sigma^<(1,2)} \quad (39)$$

This structure is similar to the one of the GF, apart from the 1st term at equal times, which is absent for the GF.

7. Let us calculate Σ^δ explicitly (see (37)):

$$\Sigma^\delta(1,2) = \langle \{ J_1, \psi_2^\dagger \} \rangle_{t_1 = t_2} = \delta_{x_1 x_2} \int d3\, v_{13} \langle \rho_3 \rangle - v_{12} \langle \psi_2^\dagger \psi_1 \rangle$$

Here

$$\langle \rho_3 \rangle = \langle \rho(x_3 t_3) \rangle = \langle \psi^\dagger_{\mathcal{H}}(3)\, \psi_{\mathcal{H}}(3) \rangle = -i\, G^<(3,3^+)$$

$$\langle \psi_2^\dagger \psi_1 \rangle = \langle \psi^\dagger_{\mathcal{H}}(2)\, \psi_{\mathcal{H}}(1) \rangle = i\, G^<(1,2)$$

$$\hookrightarrow \boxed{\begin{aligned} \Sigma^\delta(1,2) &= \underbrace{\Sigma^\delta(x_1 t_1, x_2 t_2)}_{\text{note: } t_1 = t_2 !} = -i \delta_{x_1 x_2} \int dx_3\, v(x_1, x_3)\, G^<(x_3 t_1, x_3 t_1) \\ &\quad + i\, v(x_1, x_2)\, G^<(x_1 t_1, x_2 t_1) \equiv \Sigma_{HF}(1,2) \end{aligned}} \quad (40)$$

7. <u>Dyson equations</u>

Consider GF without Σ, satisfying simply:

$$\boxed{\left(i \partial_{t_1} - h_1\right) G^0_{12} = \delta_{12}} \quad \text{and} \quad \boxed{\left(-i \partial_{t_2} - h_2\right) G^0_{12} = \delta_{12}} \quad (41)$$

Define the inverse GF via:

$$\boxed{\int G^{-1}(1,2)\,G(2,3)\,d2 = \int G(1,2)\,G^{-1}(2,3)\,d2 = \delta(1,3)} \qquad (42)$$

From (41): multiply by $G_0^{-1}(2,3)$ and $\int d2$ both sides:

$$\int d2\,\boxed{(+i\partial_{t_1} - h_1)}\,\boxed{G_0(1,2)\,G_0^{-1}(2,3)} = \int d2\,\delta(1,2)\,G_0^{-1}(2,3)$$

$$\cancel{(i\partial_{t_1} - h_1)\,\delta(1,3)} \quad \boxed{(i\partial_{t_1} - h_1)\,\delta(1,3) = G_0^{-1}(1,3)} \qquad (43)$$

and also similarly

$$\boxed{(-i\partial_{t_2} - h_2)\,\delta(1,2) = G_0^{-1}(1,2)} \qquad (44)$$

(something very similar to matrix manipulations)

- Now start from (38) and insert the unit "matrix" inside on the RHS:

$$\begin{cases} \int d3\,\boxed{(i\partial_{t_1} - h_1)\,\delta_{13}}\,G_{32} = \delta_{12} + \int d3\,\Sigma_{13}\,G_{32} \\ \int d3\,\boxed{(-i\partial_{t_2} - h_2)}\,G_{13}\,\boxed{\delta_{32}} = \delta_{12} + \int d3\,G_{13}\,\Sigma_{32} \end{cases}$$

↳ boxed quantities correspond to G_0^{-1}:

$$\begin{cases} \int d3\,G_0^{-1}(1,3)\,G(3,2) = \delta_{12} + \int d3\,\Sigma_{13}\,G_{32} \\ \int d3\,G_{13}\,G_0^{-1}(3,2) = \delta_{12} + \int d3\,G_{13}\,\Sigma_{32} \end{cases}$$

Multiply by $G_0(4,1)$ and $\int d1$ the 1st one, and $\times$ by $G_0(2,4)$ and $\int d2$ the 2nd:

$$\boxed{\begin{cases} G(4,2) = G_0(4,2) + \int d3\; G_0(4,1)\,\Sigma(1,3)\,G(3,2) \\ G(1,4) = G_0(1,4) + \int d3\; G(1,3)\,\Sigma(3,2)\,G_0(2,4) \end{cases}} \quad (45)$$

which are 2 forms of the Dyson eq.:

$$\boxed{G = G_0 + G_0 \Sigma G \;\text{ or }\; G = G_0 + G \Sigma G_0} \leftarrow \text{symbolic form} \quad (46)$$

The 2nd form also directly follows from the first:

$G = G_0 + G_0 \Sigma G$

Multiply from the ~~left~~ right by $G^{-1} \Rightarrow$ ~~$1 = G^{-1} G_0 + G^{-1} G_0 \Sigma G$~~

$\hookrightarrow 1 = G_0 G^{-1} + G_0 \Sigma$

Multiply by G_0^{-1} from the left:

$$\boxed{G_0^{-1} = G^{-1} + \Sigma} \longleftarrow \text{which is yet another form of the Dyson eq.} \quad (47)$$

Multiply by G from left and by G_0 from the right:

$G = G_0 + G \Sigma G_0$,

the other form.